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| **1.08 Integration** | | | | |
| 1.08a | Fundamental theorem of calculus | a) Know and be able to use the fundamental theorem of calculus.<br>*i.e. Learners should know that integration may be defined as the reverse of differentiation and be able to apply the result that* $\int \mathrm{f}(x)\,\mathrm{d}x = \mathrm{F}(x) + c \Leftrightarrow \mathrm{f}(x) = \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{F}(x))$, *for sufficiently well-behaved functions.*<br>*Includes understanding and being able to use the terms indefinite and definite when applied to integrals.* | | MH1 |
| 1.08b<br>1.08c | Indefinite integrals | b) Be able to integrate $x^n$ where $n \neq -1$ and related sums, differences and constant multiples.<br>*Learners should also be able to solve problems involving the evaluation of a constant of integration e.g. to find the equation of the curve through* $(-1, 2)$ *for which* $\frac{\mathrm{d}y}{\mathrm{d}x} = 2x + 1$. | c) Be able to integrate $\mathrm{e}^{kx}$, $\frac{1}{x}$, $\sin kx$, $\cos kx$ and related sums, differences and constant multiples.<br>[*Integrals of* $\arcsin$, $\arccos$ *and* $\arctan$ *will be given if required.*]<br>*This includes using trigonometric relations such as double-angle formulae to facilitate the integration of functions such as* $\cos^2 x$. | MH2 |
| 1.08d<br>1.08e<br>1.08f | Definite integrals and areas | d) Be able to evaluate definite integrals.<br>e) Be able to use a definite integral to find the area between a curve and the *x*-axis.<br>*This area is defined to be that enclosed by a curve, the x-axis and two ordinates. Areas may be included which are partly below and partly above the x-axis, or entirely below the x-axis.* | f) Be able to use a definite integral to find the area between two curves.<br>*This may include using integration to find the area of a region bounded by a curve and lines parallel to the coordinate axes, or between two curves or between a line and a curve.*<br>*This includes curves defined parametrically.* | MH3 |

| OCR Ref. | Subject Content | Stage 1 learners should ... | Stage 2 learners additionally should ... | DfE Ref. |
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| 1.08g | Integration as the limit of a sum | | g) Understand and be able to use integration as the limit of a sum.<br>*In particular, they should know that the area under a graph can be found as the limit of a sum of areas of rectangles.*<br>*See also 1.09f.* | MH4 |
| 1.08h | Integration by substitution | | h) Be able to carry out simple cases of integration by substitution.<br>*Learners should understand the relationship between this method and the chain rule.*<br>*Learners will be expected to integrate examples in the form* $\mathrm{f}'(x)(\mathrm{f}(x))^n$, *such as* $(2x+3)^5$ *or* $x(x^2+3)^7$, *either by inspection or substitution.*<br>*Learners will be expected to recognise an integrand of the form* $\frac{k\mathrm{f}'(x)}{\mathrm{f}(x)}$ *such as* $\frac{x^2+x}{2x^3+3x^2-7}$ *or* $\tan x$.<br>*Integration by substitution is limited to cases where one substitution will lead to a function which can be integrated. Substitutions may or may not be given.*<br>*Learners should be able to find a suitable substitution in integrands such as* $\frac{(4x-1)}{(2x+1)^5}$, $\sqrt{9-x^2}$ *or* $\frac{1}{1+\sqrt{x}}$. | MH5 |

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| 1.08i | Integration by parts | | i) Be able to carry out simple cases of integration by parts.<br>*Learners should understand the relationship between this method and the product rule.*<br>*Integration by parts may include more than one application of the method e.g.* $x^2 \sin x$*.*<br>*Learners will be expected to be able to apply integration by parts to the integral of* $\ln x$ *and related functions.*<br>[*Reduction formulae are excluded*.] | MH5 |
| 1.08j | Use of partial fractions in integration | | j) Be able to integrate functions using partial fractions that have linear terms in the denominator.<br>*i.e. Functions with denominators no more complicated than the forms* $(ax+b)(cx+d)^2$ *or* $(ax+b)(cx+d)(ex+f)$*.* | MH6 |
| 1.08k | Differential equations with separable variables | | k) Be able to evaluate the analytical solution of simple first order differential equations with separable variables, including finding particular solutions.<br>*Separation of variables may require factorisation involving a common factor.*<br>*Includes: finding by integration the general solution of a differential equation involving separating variables or direct integration; using a given initial condition to find a particular solution.* | MH7 |

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| 1.08l | Interpreting the solution of a differential equation | | l) Be able to interpret the solution of a differential equation in the context of solving a problem, including identifying limitations of the solution.<br>*Includes links to differential equations connected with kinematics.*<br>*e.g. If the solution of a differential equation is* $v = 20 - 20e^{-t}$*, where* $v$ *is the velocity of a parachutist, describe the motion of the parachutist.* | MH8 |